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A proficiency testing of electromagnetic radiated emission measurements

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Abstract
A proficiency testing is carried out for electromagnetic radiated emission (RE) measurements in the frequency range of 30–1000 MHz. A spherical dipole radiator (SDR) is employed as a reference emitting source. Using the SDR, the RE measurements are taken for horizontal and vertical polarizations at 12 open area test sites and the measurement distance is 10 m. To evaluate the proficiency of each test laboratory, the robust Z score and E_n ratio methods are used. In order to use the E_n ratio method, the measurement uncertainty of each test laboratory is evaluated in accordance with the International Organization for Standardization guide to the expression of uncertainty in measurement. Results show that decision on a proficiency testing depends on the evaluation method used.

Keywords: proficiency testing, radiated emission, open area test site, reference emitting source, measurement uncertainty

1. Introduction

Assessing the competence of accredited laboratories has become more and more important in the metrology and testing community. One assessment technique commonly used is proficiency testing, which involves the determination of laboratory performance by means of interlaboratory comparisons, whereby the laboratory undergoes practical tests and their results are compared with those of other laboratories. Several organizations have proposed and operated the control program and evaluation method of a proficiency testing system [1–3]. According to the National Association of Testing Authorities (NATA) guide, Australia [1], the Z score method is to be used for 'test' laboratories and the E_n ratio method is employed to 'calibrate' laboratories. However, provided the reference value (RV) and uncertainty of each test laboratory are known, the latter method can also be applied to test laboratories. The RV is defined as an assigned value which is provided by a reference laboratory. However, the RV presented in this paper will not be given by a reference laboratory but obtained by averaging the data of the participant test laboratories.

Electromagnetic (EM) emission is the phenomenon by which EM energy emanates from a source. The EM energy of a source can be guided by conductors such as signal cables and their shields, pipes used to bring various services inside enclosures, and other structural parts or the energy can propagate in the form of EM waves. We refer to the former as conducted emission and to the latter as radiated emission (RE). The EM energy from the source may degrade the performance of a device, piece of equipment or system, which is referred to as electromagnetic interference (EMI).

Many articles dealing with the repeatability of RE measurements have appeared [4–9] where the RE of electrical and electronic equipment is described as an electric field strength measured at an open area test site (OATS) or alternative test facilities. An ideal OATS consists of a perfectly conducting ground plane of infinite extent and infinite half-space free from all obstructions over the ground plane. The mean and standard deviation have been generally used to present the repeatability in RE measurements. The test competence of 13 laboratories was assessed by measuring RE at 10 m distance and the measurements were

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made at OATS, a semianechoic chamber and a gigahertz transverse electromagnetic (GTEM) cell using a commercial pulse/function generator as a reference emitting source [8]. In [8], proficiency testing was presented in terms of the mean of the data deviation, standard deviation and safety factor, defined as the mean of the data deviation plus twice the standard deviation.

In this paper, proficiency testing of EM RE measurements is performed in the frequency range of 30–1000 MHz. A commercially available spherical dipole radiator (SDR) with a repeatability of 0.2–0.5 dB is employed as a reference emitting source. The measurand in RE measurements is the electric field strength in units of V m^{-1} or dBμV m^{-1} and it will be referred to as the 'field strength'. Measurements are made at 12 OATSs for horizontal and vertical polarizations and the measurement distance is 10 m. Only the amplitude of the field strength is of interest for the proficiency testing presented in this paper since the frequency to be measured is known to the test laboratories in advance. We use two evaluation methods to make a decision on the proficiency of a test laboratory: the robust Z score and E_n ratio methods which will be discussed in the following section. For the E_n ratio method, the measurement uncertainty is evaluated in accordance with the International Organization for Standardization (ISO) guide to the expression of uncertainty in measurement [10].

2. Proficiency evaluation methods

In this paper, we employ two evaluation methods for proficiency testing. Firstly, the robust Z score method is based on the median and normalized interquartile range (IQR). For a given data set, the IQR is the difference between the lower and upper quartiles. The lower quartile $Q1$ is the value below which, as near as possible, a quarter of the results lie. Similarly the upper quartile $Q3$ is the value above which a quarter of the results lie. In most cases, $Q1$ and $Q3$ are obtained by interpolating between the data values and thus IQR $= Q3 - Q1$. The Z score for a result is given by

$$Z = \frac{x - X}{0.7413 \times \mathrm{IQR}} \tag{1}$$

where x and X are the result of a test laboratory and the median of all results, respectively. The denominator in (1) is the normalized IQR where the factor 0.7413 comes from the normal distribution with a mean of zero and a standard deviation of one. The decision criteria of this method is $|Z| \leqslant 2$ for satisfactory, $2 < |Z| < 3$ for questionable or encouraging to take a closer look at the result, and $|Z| \geqslant 3$ for unsatisfactory [1, 3].

Next, the E_n ratio stands for error normalized and is defined as

$$E_n = \frac{x - X}{\sqrt{U_x^2 + U_X^2}} \tag{2}$$

where x and X are a test laboratory's result and the RV, respectively. For a result to be satisfactory, $|E_n| < 1$. The U_x is the uncertainty (generally $k = 2$, where k is the coverage factor which provides an interval having a specified level of confidence) including instrumentation, receiving antenna and the OATS itself for each laboratory and is evaluated according to the ISO guide [10]. Since there is no ideally perfect OATS, the best alternative to the RV is the arithmetic mean of the measured electric field strengths [5, 7]. In equation (2), U_X is the uncertainty of the mean, which will be explained in the relevant section.

An observation on (1) and (2) is that the Z score method is more stringent than the E_n ratio in view of the denominators of the two equations, which can be considered as a weighting factor. This is because in the former method the deviation is weighted by only the normalized IQR corresponding to the conventional standard deviation, whereas in the latter the deviation is weighted by the measurement uncertainty of a test laboratory as well as the standard deviation of the mean.

3. Measuring electric field strengths

3.1. The reference emitting source

In the RE proficiency testing, a commercially available SDR, SRS2100, Seth Corporation, USA, with a repeatability of 0.2–0.5 dB has been used as a reference emitting source. Its frequency range is 10 MHz–2.5 GHz. The SDR system consists of a spherical radiator with a 10 cm outer diameter and a control unit to interface to the operator and an external rf signal source. The radiator is a transmitting antenna whose radiation pattern is explicitly known and electrically isolated from the rf source by optical fibre cables. The use of optical fibre cables between the control unit and the radiator eliminates the disruption of the radiated field to be measured. More details on the SDR can be found in [11].

Before proficiency testing commences, the rf input power for each frequency is determined in advance at the pilot site so that a subsequent gap voltage of the SDR is in a typical useful range of 0.2–1.0 V. The gap voltage is displayed on a display panel of the controller and it monitors the rf voltage between the gap of two hemispherical radiating elements in a dc voltage. The gap voltages vary in the useful range because of the characteristic of the amplifiers. During the procedure of determining the gap voltage, care should be taken not to overdrive the input of the rf amplifier installed in the spherical radiator. The gap voltage determined above is referred to as a reference gap voltage. For the repeatability of the rf field generation, the gap voltage monitored at each site should be maintained as close as possible to the reference voltage at each frequency during proficiency testing. The gap voltage of the SDR has been monitored within +12/−5 and +36/−22 mV with respect to the reference gap voltage for horizontal and vertical polarizations, respectively. For the vertical polarized case, the maximum variation of the gap voltages monitored at the 12 sites is +36 mV at the frequency of 150 MHz and the corresponding reference gap voltage is 0.828 V. Since the rf amplifier installed in the spherical radiator operates linearly with respect to the rf input power, the variation in the electric field at each site will be $20 \log \frac{0.828\ \mathrm{V}+0.036\ \mathrm{V}}{0.828\ \mathrm{V}} = 0.4$ dB. Moreover, since the gap voltage variation for horizontal polarization is smaller than the one for vertical polarization as presented earlier, the resultant repeatability of site-to-site field generation is within 0.4 dB.

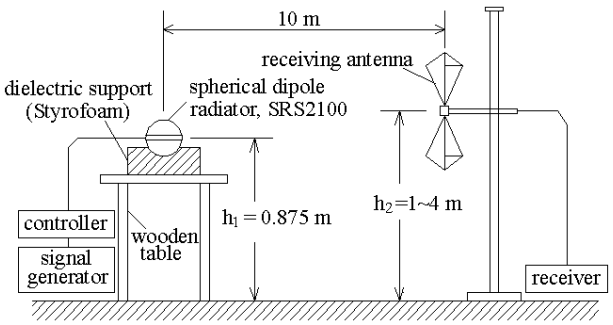


Figure 1. Set-up of RE measurements.

3.2. The measurement set-up

At each test laboratory the RE measurements have been made according to international standards such as ANSI C63.5 [12] and CISPR 16-2 [13]. The measurement set-up is illustrated in figure 1. The SDR is positioned at the centre of a turntable on a 0.8 m high wooden table. A styrofoam support is used to place the spherical radiating element on the table and the height of the centre of the SDR is 0.875 m above the ground plane. Measurements are made at a distance of 10 m for both horizontal and vertical polarizations. Since the electric field at an OATS consists of direct and ground-reflected waves, the field strength at an observation point varies with the receiving antenna height. According to international standards, the receiving antenna is scanned 1–4 m at a 10 m distance and maximum field strengths are recorded [12]. The receiving antenna can be a set of biconical and log-periodic antennas, or a bilog antenna, to cover the full frequency range of interest. An EMI receiver capable of measuring the electric field strengths is used in the quasi-peak mode with 120 kHz bandwidth.

The EMI receiver gives the measuring quantity in terms of rf power or voltage. In order to convert the measured rf power into a field strength the following relation is used:

$$E = P_r + L - G + 107.0 + AF_R \qquad (3)$$

where E is the electric field strength in dBμV m^{-1}, P_r is the received power measured by the 50 Ω EMI receiver in dBm, L is the cable loss in dB and G is the gain of the preamplifier in dB.

AF_R the antenna factor (AF) of the receiving antenna provided by the manufacturer, dB(m^{-1}).

In equation (3), the factor 107 dB is the conversion factor of the received power in units of dBm measured by a 50 Ω receiver into the voltage in units of dBμV. The AF of an antenna is defined as the ratio of the incident electric field at the surface of the measurement antenna to the received voltage at the antenna terminal. To measure the received voltage, an EMI receiver is attached to the terminals of the measurement antenna by rf cables. Since the manufacturer's AF has been calibrated for free-space conditions, the AF for an OATS is different from the manufacturer's data because of the interaction between the antenna and the ground plane. Although the AF of an antenna depends heavily on the height above the ground plane and the separation distance [14, 15], we simply use the manufacturer's AF calibrated at a distance of 10 m .

In proficiency testing, the SDR system has been carried from site to site to make on-site measurements. Since the spherical dipole antenna with a uniform excitation voltage at the equator radiates as a dipole antenna, the rotation of the turntable is not necessary. Because of the limited duration of the battery for the SDR, the measurement of the selected frequencies has been made for approximately 2 h at each site. Eleven test laboratories participated in the proficiency testing performed in the second half of 1999 and all laboratories are accredited by the registration scheme for EMI testing laboratories under the supervision of the Ministry of Information and Communication, Republic of Korea. Since one laboratory has two sites, denoted as 'B1' and 'B2' in the following sections, and the laboratory uses two independent measurement systems for the two sites, the total number of sites is 12.

3.3. Measurement results

Measurements have been made at 23 discrete frequencies selected arbitrarily from the frequency list of ANSI C63.4 [12]; 30, 35–50, 60–100, 125–200, 250, 300, 400–1000. At several test sites one or two test frequencies were skipped because of high external ambient noise, e.g. frequency modulation (FM) and television (TV) broadcast signals.

Figures 2(a) and (b) illustrate the measured electric field strengths for horizontal and vertical polarizations, respectively. It is noted that the number of measured field strengths is 10–12 at all frequencies except at 100 MHz. Because of ambient noise the number of measured field strengths at 100 MHz is 5. The relatively large uncertainty of the RV at the frequency 100 MHz in figure 2 is attributed to the small amount of data at the frequency (refer to section 4.2). The uncertainty U_X of the RV ranges from 0.6–2.1 dB for horizontal polarization and from 0.7–3.2 dB for vertical polarization. The evaluated expanded uncertainty U_x varies with the instrumentation of each laboratory. As an example, the evaluated expanded uncertainty U_x ($k = 2$) of laboratory 'A' is shown by the dotted curve in figure 2. The expanded uncertainty of the laboratories is +5.6/−5.4 and +5.0/−4.4 dB for the frequency ranges of biconical and log-periodic antennas, respectively. The AF of an antenna depends on the height and responses of an antenna in the direction of the direct ray and in the direction of the ground-reflected ray vary because of the antenna directivity; the effect of an unbalanced antenna is greatest when the input coaxial cable is aligned parallel to the antenna elements. The degree of dependence of the three above-mentioned phenomena varies with polarization. Subsequently, the uncertainty also depends on the polarization, but not significantly. Note that the uncertainty U_x and U_X is required for the E_n ratio method, whereas evaluating the uncertainty is not needed for the Z score method.

3.4. Measurement uncertainty

The uncertainty in RE measurements has been considered in many articles [9, 16–18]. Since we focus on proficiency testing itself, we do not discuss measurement uncertainty in detail. Instead, we only list the contributions to the uncertainty in RE measurements and more details can be found in [9]. It should

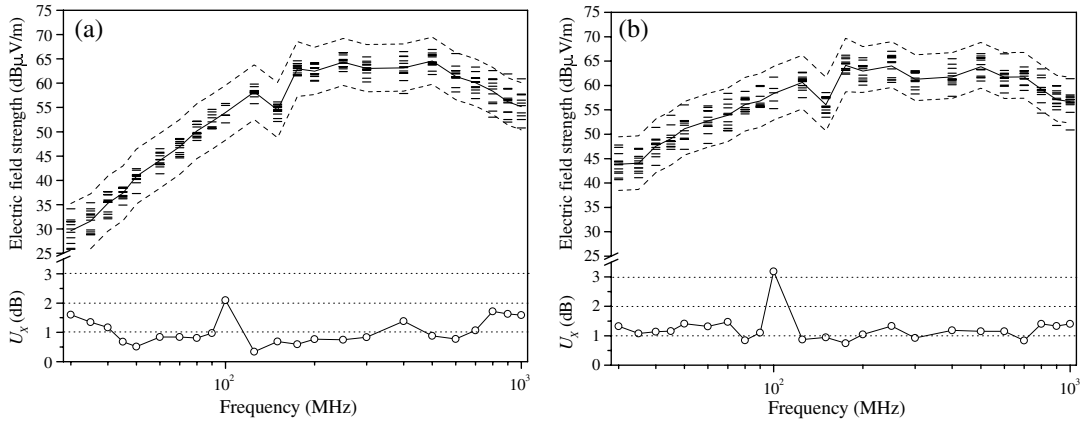


Figure 2. Measured electric field strengths (a) for horizontal and (b) for vertical polarizations. —— X is the RV (arithmetic mean) and - - - - U_x is the expanded uncertainty ($k = 2$) in RE measurements of laboratory 'A', which varies with the instrumentation of a test laboratory.

be emphasized that the uncertainty of the test laboratories has been evaluated in accordance with the ISO guide [10].

To evaluate uncertainty, contributions to uncertainty are grouped into three parts: the transmitting and receiving parts, and the OATS itself.

- Transmitting part: the absolute level accuracy of a signal source, repeatability of the SDR itself, mismatch between a signal source and the SDR and measurement distance variation. It is to be noted that, for actual RE tests, the emitter is generally unknown and the uncertainty due to the transmitting part is not accounted for.
- Receiving part: uncertainty in the AF calibration, AF variation with height, AF interpolation versus frequency, antenna directivity, antenna phase centre variation, mismatch between a receiving antenna and a receiver and receiver specification including voltage reading accuracy, absolute pulse response and pulse repetition response. Since a continuous-wave (cw) signal is applied to the reference emitting source, the SDR, both the absolute pulse response and pulse repetition response do not contribute to the uncertainty. However, for actual RE tests, an appropriate tolerance for the two pulse responses should be assigned.
- OATS imperfection: the RE measurements have been made at ten 10 m OATSs and at two 30 m OATSs. The size of the ground plane of the 10 m OATSs is (18–23 m)(L) × (8–18 m)(W). For one of the two 30 m OATSs, the ground plane of the 10 m distance measurement is 22 m(L) × 18 m(W) and for the 30 m distance measurement a narrower wire mesh of 38 m(L) × 8 m(W) is extended along the measurement axis. The other 30 m OATS has the ground plane of 60 m(L) × 52 m(W). The ground plane of all sites consists of metal plates and galvanized wire mesh with a spacing of less than 5.0 mm. The flatness of all sites is satisfied by the Rayleigh criterion, which provides a useful estimate of maximum allowable ground plane roughness [19]. The deviation of normalized site attenuation from the theoretical values [12, 13] for the twelve sites are within +1.7 ∼ +3.7/–3.3 ∼ −2.2 dB.

In order to evaluate the expanded uncertainty of each test site the frequency range of 30–1000 MHz is divided into two ranges, corresponding to those of the receiving antenna. In general, the receiving antenna is composed of biconical and log-periodic antennas. The associated transition frequency between the two antennas is either 200 or 300 MHz and the field strength measured by the log-periodic antenna is chosen at the transition frequency. A bilog antenna for the frequency range of 30–1000 MHz is also used at several test sites. The voltage standing wave ratio (VSWR) of the bilog antenna deteriorates as the frequency becomes low. If the maximum VSWR is chosen in the full operating frequency range of the bilog antenna, the uncertainty due to mismatch in its relatively high frequency range will be unnecessarily overestimated. For such cases, the transition frequency is arbitrarily set to be 300 MHz to evaluate the uncertainty.

The evaluated uncertainty for each site depends on its instrumentation and the resultant expanded uncertainty ($k = 2$) is within +4.7 ∼ +5.6/–5.7 ∼ −5.0 dB and +4.2 ∼ +4.9/–4.8 ∼ −4.2 dB for the frequency ranges of biconical and log-periodic antennas, respectively.

4. Results of the proficiency testing

We evaluated the proficiency of the test laboratories using the robust Z score and E_n ratio methods.

4.1. Using the robust Z score method

As observed in equation (1), the Z score is straightforwardly evaluated. The median and normalized IQR are measures of the centre and the spread of the data, respectively, and they are robust statistics, which means that they are not influenced by the presence of outliers in the data. Figure 3 shows the evaluated Z scores for horizontal (a) and vertical (b) polarizations. It is observed in figure 3 that the Z score of laboratory 'K' has a negative value at many frequencies. This means that the laboratory has a systematic problem with its measuring procedure that needs to be resolved. The questionable frequencies ($2 < |Z| < 3$) as well as outlying frequencies ($|Z| \geqslant 3$) of the laboratories are summarized in table 1. Here, the term 'questionable' means that laboratories

Table 1. Outlying frequencies in units of MHz determined by the Z score and E_n ratio methods in RE measurements.

	Horizontal polarization					Vertical polarization				
	Robust Z score				E_n number	Robust Z score				E_n number
Lab. id.	$2 < \|Z\| < 3$		$\|Z\| \geqslant 3$		$\|E_n\| \geqslant 1$	$2 < \|Z\| < 3$		$\|Z\| \geqslant 3$		$\|E_n\| \geqslant 1$
A						90				
B1	125					400	1000	90		
B2						800				
C										
D	400	700	600		400					
E	50		125			30	45			
						400	800			
F	700					400		800		
G	300	900			900	40		1000		1000
	1000				1000					
H	300				800	800				
I	50	150	125	600		125		40	600	
J								45		
K	45	125	50			40	600	800	1000	900
	150	300				700	900			1000
	600									

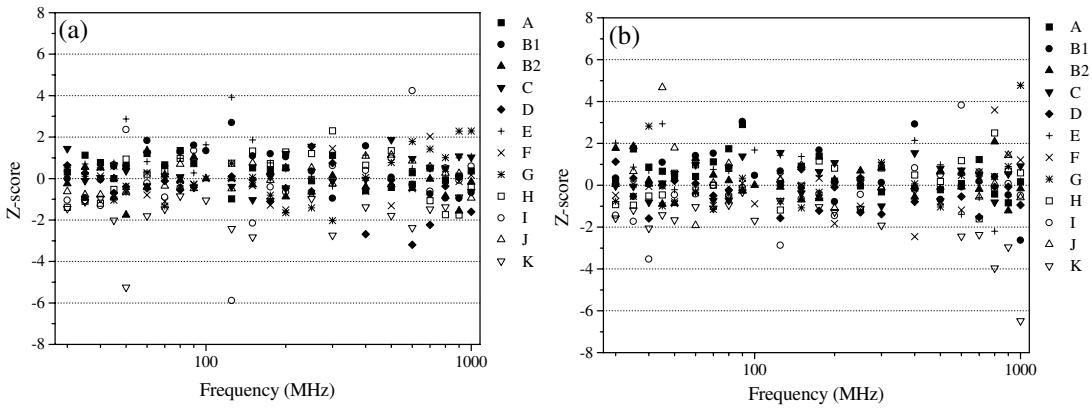


Figure 3. Evaluated Z scores for the test laboratories for (a) horizontal and (b) vertical polarization, respectively.

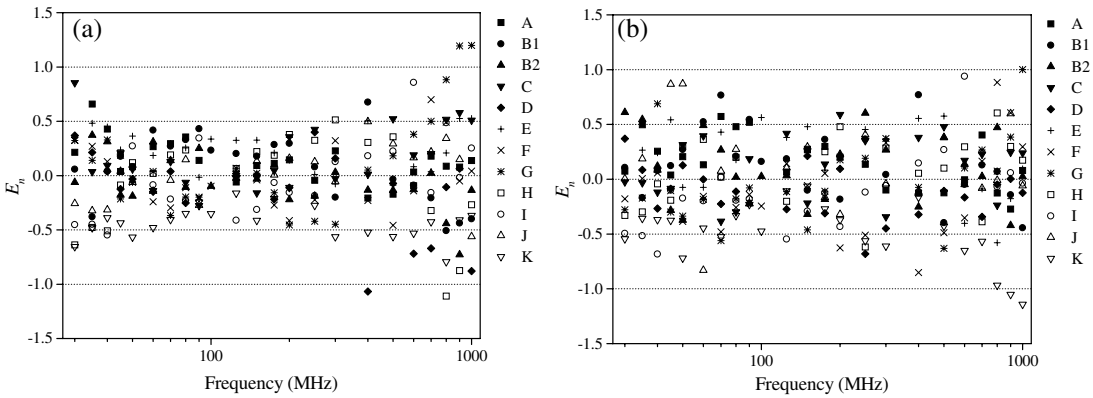


Figure 4. Evaluated E_n ratios of the test laboratories for (a) horizontal and (b) vertical polarizations, respectively.

which have a Z score in this region ($2 < |Z| < 3$) are encouraged to take a close look at their results.

4.2. Using the E_n ratio method

In the E_n ratio method as described in equation (2), it is essential to evaluate the uncertainty of the RV as well as the measurement uncertainty of a test laboratory. As stated earlier, the arithmetic mean at each frequency serves as the RV.

In order to define the uncertainty of the RV, the electric field strengths measured at each site are considered as those obtained from repeated measurements. We define the standard uncertainty for the repeated measurements as follows. With n-independent single measurements, the estimated value of x_i

Table 2. Multiplying factors.

No. of measurements	2	3	4	5	6	7	8	9
Multiplying factor	6.99	2.27	1.66	1.44	1.33	1.26	1.22	1.19

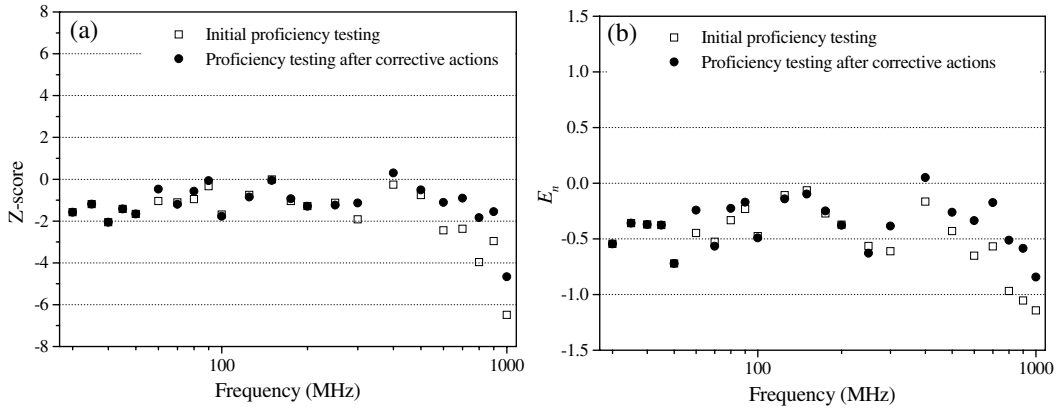


Figure 5. The (a) Z scores and (b) E_n ratios of laboratory 'K' for vertical polarization. The results show that the proficiency of the laboratory has been improved by corrective actions.

of the true value of X_i is given by the arithmetic mean of the individual measured values $v_{i,j}$ $(j = 1, \ldots, n)$:

$$x_i = \bar{v_i} = \frac{1}{n}\sum_{j=1}^{n} v_{i,j} \tag{4}$$

where $v_{i,j}$ denotes the field strength of the ith frequency at the jth test site. The estimated standard deviation s_{v_i} of the distribution of the measured values v_i is

$$s_{v_i} = u(v_i) = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(v_{i,j} - \bar{v_i})^2}. \tag{5}$$

When repeated measurements are made, the standard deviation of the mean s_{x_i} is

$$s_{x_i} = u(x_i) = \frac{s_{v_i}}{\sqrt{n}} = \sqrt{\frac{1}{n(n-1)}\sum_{j=1}^{n}(v_{i,j} - \bar{v_i})^2}. \tag{6}$$

Here, $2s_{x_i}$ is referred to as the standard uncertainty of the mean U_X at the ith frequency. Equation (6) is exactly true only if n is infinite. In practice, most standards laboratories either use $n \geqslant 10$ or apply the appropriate multiplying factors given in table 2 [20]. The number of field strength data n collected at each frequency is shown in table 3. In other words, n indicates the number of test sites at which the field strength was measured at the frequency without high ambient noise.

Figure 4 illustrates the evaluated E_n ratio of the test laboratories. The outlying frequencies at which the evaluated E_n ratio meets $|E_n| \geqslant 1$ are listed in table 1 for each laboratory. As observed in table 1, the outlying frequency determined by the robust Z score and E_n ratio methods is quite different. However, excluding laboratory 'H', all outlying frequencies at which the evaluated E_n ratio meets $|E_n| \geqslant 1$ also meets the condition $|Z| > 2$, but not vice versa.

Compare the two methods of proficiency testing by means of the results shown in table 1. For each laboratory, outlying frequencies depend on the evaluation method. Considering the results of table 1, laboratories can be categorized into four groups as follows. The outlying frequencies determined by the two evaluation methods were accounted for as outliers for categorization.

Table 3. The number of field strength data n at the measurement frequencies.

Horizontal polarization	Vertical polarization
5; 100 MHz	5; 100 MHz
10; 90 MHz	10; 90 MHz
11; 30 MHz	12; elsewhere
12; elsewhere	

- Group 1: laboratories 'A', 'B2' and 'C'. They have no outlying frequencies and show fairly good proficiency.
- Group 2: laboratories 'B1', 'E', 'F', 'I' and 'J'. These laboratories have outlying frequencies determined by the robust Z score method only. All laboratories except laboratory 'I' have only one outlying frequency. A notice that attention should be paid during the measurement at the outlying frequency was included in the report for the laboratories of this group.
- Group 3: laboratory 'H'. It has one outlier for the E_n ratio method only. For this laboratory, careful examination and corrective action should be carried out for the test system.
- Group 4: laboratories 'D', 'G' and 'K'. These laboratories have outlying frequencies for both proficiency evaluation methods. For these three laboratories, careful examination and corrective action should be carried out for their test system. It may include the calibration of the antennas and EMI receiver, review of the measurement software, qualification of the OATS, etc. For example, laboratory 'K' recalibrated the AF of antennas used and results for vertical polarization are shown in figure 5. After the AF was recalibrated, the proficiency of laboratory 'K' has been improved noticeably, as observed in figure 5. The

calibration of AF is the corrective action for the laboratory. However, it is noted that there is still an outlying frequency for the Z score method in figure 5(a) while there is no outlier for the E_n ratio method in figure 5(b).

In conclusion, the discrepancy in the outlying frequencies determined by the two methods stems mainly from the fact that the median in the Z score method and the RV, the arithmetic mean, in the E_n ratio method are different in principle. According to [1], the E_n ratio method is for 'calibration laboratories'. Nevertheless, we attempted to apply the method to 'test' laboratories because the method accesses the overall capability of a laboratory, i.e. the closeness of the results to the RV and measurement uncertainty. In our viewpoint, the results presented in this paper can serve as useful resources for the relevant organizations in assessing the competence of test laboratories.

5. Conclusion

We performed proficiency testing of EM RE measurements using the Z score and E_n ratio methods in the frequency range of 30–1000 MHz. The SDR has been employed as a reference emitting source. Using the SDR, measurements are taken for horizontal and vertical polarizations at 12 OATSs and the measurement distance is 10 m.

For field strengths measured at each test laboratory, the Z score has been evaluated in a straightforward way, while the E_n ratio has been obtained through the evaluation of the measurement uncertainty of each laboratory and the uncertainty of the RV. By applying appropriate criteria for the two evaluation methods, the outlying frequencies have been determined. Results show that the decision on proficiency testing of a laboratory depends on the evaluation method used. Nevertheless, the results presented in this paper should be useful to those organizations tasked with assessing the competence of test laboratories.

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